Theory

Abstract

We specify objective and admissibility definitions for counterfactual evaluation under deterministic replay, and state the invariants required for unit-consistent multi-metric regret attribution. The document is intended as a constitutive specification: implementations are correct only insofar as they preserve causal closure, dimensional consistency, and admissible comparators.

1 Governing Definitions and Scope

This document specifies the objective, admissibility constraints, and invariants for the Gedanken Engine for regret minimization. Implementations are considered correct only insofar as they satisfy these definitions, independent of empirical performance.

Governing relations. We assume evaluation is a closed, replayable system in which outcomes are deterministic functions of the boundary conditions and the alternative. Closure holds if and only if identical (s, a) pairs produce identical observations and metrics under replay.

$$M : (s, a) \mapsto M(s, a). \tag{1}$$

Counterfactual comparisons are defined only when the boundary conditions are identical across alternatives (causal closure): (s, a) and (s, a') are comparable only if replay begins from an identical boundary condition s.

$$(s, a) \sim (s, a') \;\Rightarrow\; s \text{ is identical under replay.} \tag{2}$$

Scope (hard requirements).

- **Offline measurement:** metric computation and regret attribution are performed post-execution.
- **Replayability:** for fixed `(spec, seed, trace)`, the execution is a pure function and must be reproducible.
- **No partial resets:** if an external dependency cannot be reset to the initial boundary condition, the counterfactual is undefined.
- **Unit consistency:** any aggregate regret scalar over multiple metrics must be dimensionless via explicit normalization. Each metric must declare its physical units and its characteristic scale prior to aggregation.

Out of scope. Algorithmic design choices (search, learning, and control strategies) are not prescribed here; only the quantities to be measured and the invariants they must satisfy are defined.

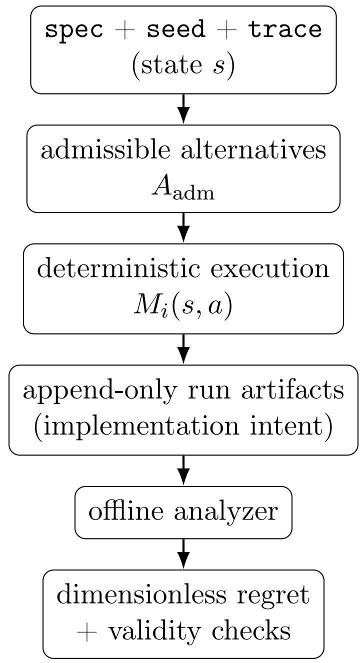

Figure 1: Evaluation pipeline from frozen boundary conditions to offline regret attribution.

2 Problem Formulation

We model evaluation as a closed, replayable system.

- **State** s: complete boundary conditions for evaluation, including immutable workload traces, environment reset state, RNG seed state, and all other factors that causally influence state transitions, observations, or metric realizations.
- **Alternative** $a \in A$: a finite, enumerated set of admissible actions, policies, or designs.
- **Outcome** $M(s, a)$: deterministic mapping from (s, a) to observed metrics (vector-valued).

2.1 Evaluation Pipeline (Overview)

2.2 Closed-System Assumption (Causal Closure)

All counterfactual evaluations are performed against the same boundary conditions.

If any external dependency or environmental influence cannot be reset, mocked, or deterministically replayed as part of the state s, the counterfactual is undefined. In this case, the evaluation must terminate without emitting regret values.

3 Regret

Regret is counterfactual loss relative to the best admissible alternative, evaluated under identical boundary conditions.

3.1 Scalar Regret (Single Metric)

Let $J(s, a)$ be a scalar objective to maximize. Let a^* be the best admissible alternative as defined in Section 3.3.

$$\text{Regret}(a \mid s) = J(s, a^*) - J(s, a) \tag{3}$$

3.2 Multi-Metric Regret Requires Nondimensionalization

In engineering systems, metrics have different physical dimensions (latency, throughput, energy, error rate). Direct weighted summation of heterogeneous units is dimensionally invalid and produces scale bias.

Define metric components:

- $M_i(s, a)$ for $i = 1, \ldots, k$, each with declared physical units and an associated characteristic scale $C_i(s)$ sharing the same units.

Define a **characteristic scale** $C_i(s)$ with the same units as M_i. Both the units and the scale definition are mandatory inputs to the evaluation and must be specified prior to execution. The normalization transform is:

$$r_i(s, a) = \frac{M_i(s, a^*) - M_i(s, a)}{C_i(s)} \tag{4}$$

Then define dimensionless aggregate regret:

$$\text{Regret}(a \mid s) = \sum_{i=1}^{k} w_i \, r_i(s, a) \quad \text{(defined only if all } r_i \text{ are dimensionless).} \tag{5}$$

Constraints.

- $w_i \geq 0$
- $\sum_i w_i = 1$ is recommended for interpretability.

ASCII form.

```
r_i(s,a) = (M_i(s,a*) - M_i(s,a)) / C_i(s)
Regret(a|s) = sum_i w_i * r_i(s,a)
```

3.2.1 Recommended Choices for $C_i(s)$

Pick one and document it per metric:

1. **Range scale (robust default)**

$$C_i(s) = \max_{a \in A_{\text{adm}}} M_i(s, a) - \min_{a \in A_{\text{adm}}} M_i(s, a) \tag{6}$$

2. **Tolerance scale (SLO/SLA anchored)**

$$C_i(s) = \text{SLO}_i \tag{7}$$

(for "must not exceed" constraints)

3. **Utopia (fraction-of-optimum, use with care)**

$$C_i(s) = |M_i(s, a^*)| \tag{8}$$

(singular when optimum is 0)

This mirrors nondimensionalization practice in physics codes where equations are commonly expressed in nondimensional form to preserve invariance under unit choice.

3.3 Comparator Definition: Best Admissible Alternative

The comparator must be **admissible**: it cannot depend on information unavailable to the evaluated agent at decision time.

Let $I(h)$ be the information set induced by observable history h, parameterized by an explicit observation horizon that defines which variables, time steps, and signals are visible.

An alternative a is admissible if it maps a declared information set $I(h)$ to decisions without using hidden state (private variables, future RNG outcomes, or external oracle information), and if its declared observation horizon does not exceed that of the evaluated agent.

$$a \in A_{\text{adm}} \iff a = \pi(I(h)) \tag{9}$$

Then:

$$a^* = \arg\max_{a \in A_{\text{adm}}} J(s, a) \quad \text{subject to } \mathcal{O}(a) \preceq \mathcal{O}(a_{\text{eval}}) \tag{10}$$

This prevents "clairvoyance regret": penalizing agents for not using information outside their declared observation horizon. Such comparisons are invalid counterfactuals and are treated as correctness failures.

4 Scalarization Limits and Pareto Structure

The weighted-sum scalarization restricts optimization to the Pareto frontier. In non-convex tradeoff surfaces, some Pareto-optimal points are not representable by any fixed weight vector.

This is a documented limitation. If non-convex Pareto coverage is required, use a different scalarization (for example Chebyshev / max-norm formulations) and treat it as a separate objective definition.

5 Conservation Laws (Correctness Invariants)

5.1 Conservation of History (Deterministic Replay)

With fixed `(spec, seed, trace)` and complete state s, the execution is a pure function and produces bitwise-identical outputs:

- identical derived metrics
- identical decision sequence
- identical derived-state hash

Any divergence indicates entropy injection (race conditions, nondeterministic iteration order, unseeded randomness, external time, network I/O).

5.2 Conservation of Trace (Boundary Condition Invariance)

The workload trace is immutable and independent of the alternative being evaluated.

If choosing a changes the subsequent trace arrival pattern and that feedback is not modeled deterministically inside the frozen system, counterfactual comparison is invalid.

5.3 Conservation of Dimensionality

All aggregated regret scalars must be dimensionless.

Implementations must reject any configuration that aggregates metrics with missing unit annotations, missing characteristic scales, or heterogeneous units without explicit $C_i(s)$ normalization. Such cases are treated as correctness failures and must abort evaluation.

5.4 Offline Isolation

Metric computation and regret attribution occur post-execution.

No gradient updates or online learning occurs during measurement.

6 Validity Checks and Hard Failures

The following are treated as correctness failures, not "bad predictive regret" after admissible maximization:

- **Trace divergence** between alternatives: indicates invalid counterfactual evaluation.
- **Unit or scale violation:** missing unit annotations, missing characteristic scales, or aggregation across heterogeneous units without normalization, indicating dimensional inconsistency.
- **Oracle violation:** use of a comparator whose declared observation horizon strictly exceeds that of the evaluated agent, indicating inadmissible clairvoyance.

7 Artifact Mapping (Theory $\rightarrow$ Repo)

- s (boundary conditions): `traces/` plus frozen environment state
- A and admissibility constraints: `specs/` (policies, action sets, declared observation model and observation horizon)
- M_i: replay-derived metrics in `runs/` or `traces/` append-only outputs
- C_i: characteristic scales, defined in spec and versioned with evaluation
- Regret: evaluation artifact emitted by offline analyzer

8 Minimal Falsification Scenario (Asymmetric Information Externality)

A minimal scenario that falsifies naive regret engines:

- Two agents
- One decision step
- Two actions
- Asymmetric private information
- Externality cost

Naive engines define a^* using hidden state and measure "luck regret".

Correct engines define a^* over admissible strategies and measure "decision regret".